

Crossfade: The Mathematics to Check

A sanity-check edition for the signals lead

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Abstract

This document isolates every mathematical claim that the Crossfade platform relies on, in the order the platform relies on them: the channel representations and when each is sparse; the descriptors as functionals of a named representation; the dimensionless regime coordinates and the overlap result that falls out of them; the two simple generators and what they omit; the identifiability limits that no estimator, classical or learned, can remove; and the definition of transfer as a measured effect. Each section ends with numbered *Check* items. We are asking for three kinds of response to each: it is right, it is wrong in this way, or it is convention-dependent and here is the convention we should declare. Nothing about software is here; the full thesis is a separate document. The two items on the critical path are Check 3 (the first descriptor) and Check 6 (the first generator).

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Why these are the claims

The platform is built around one idea from the original conversation: the time-varying impulse response or scattering function of the channel is the common thread across sonar, HF skywave and RF, and every display anyone looks at is a functional of it. Two rounds of review turned that idea into a set of statements with different standing. Some are theorems about linear operators (Section 2). Some are definitions we chose and could have chosen otherwise (Sections 3 and 4). Some are modelling simplifications whose adequacy only a practitioner can judge (Section 5). Some are limits that hold regardless of method (Section 6). And some are hypotheses that the benchmark exists to measure (Section 7). The tags **[G]**, **[P]**, **[H]** say which is which, so that a wrong claim can be fixed at the right level: a bug, a swapped convention, or a recorded result.

1 Notation and conventions

Symbol	Meaning	Units
$x(t), y(t)$	transmitted and received signal (complex baseband unless stated)	—
$s(t)$	source waveform; $s_a(t)$ its analytic signal	—
$h(t, \tau)$	time-varying impulse response (Bello)	s^{-1}
$S_H(\tau, \nu)$	delay–Doppler spreading function	—
$h_W(\tau, \alpha)$	delay–scale spreading function	—
$C_H(\tau, \nu)$	scattering function, $\mathbb{E} S_H ^2$ under WSSUS	—
$P(\tau)$	power–delay profile (PDP)	—
f_c, B, T	carrier, bandwidth, observation length	Hz, Hz, s
c	propagation speed in the medium	m/s
v	radial relative speed, positive when closing	m/s
$M = v/c$	Mach number	1
$\beta_B = B/f_c$	fractional bandwidth	1
σ_τ, σ_ν	RMS delay spread, RMS Doppler spread	s, Hz
$\Pi_\tau, \Pi_\nu, \Pi_L, \Pi_f, \Pi_h, \Pi_s$	regime groups, Section 4	1
$\{(\tau_k, a_k)\}_{k=1}^K$	arrival list: delays and complex gains	s, 1
D, z_s, z_r, r	water depth, source depth, receiver depth, range	m
c_b, ρ_b	bottom half-space sound speed and density	m/s, kg/m ³

Fourier transform convention: $X(f) = \int x(t) e^{-j2\pi ft} dt$. Scale $\alpha > 1$ means time compression (a closing source). Every equation is tagged: **[G]** for something the software enforces, **[P]** for

a convention or modelling choice that can be swapped without changing a contract, **[H]** for a scientific claim the benchmark measures. Items marked *Check* are the ones we most want the reader to confirm, correct, or replace.

2 The channel and its representations

2.1 Linear time-varying channel

A single-input single-output propagation channel is modelled as a linear time-varying (LTV) operator $\mathcal{H}$ with kernel $h(t, \tau)$ [2]:

$$y(t) = (\mathcal{H}x)(t) = \int h(t, \tau) x(t - \tau) d\tau. \quad (1)$$

$h(t, \tau)$ is the response at time t to an impulse launched at time $t - \tau$. Its Fourier transform in t is the *delay–Doppler spreading function*,

$$S_H(\tau, \nu) = \int h(t, \tau) e^{-j2\pi\nu t} dt, \quad y(t) = \iint S_H(\tau, \nu) x(t - \tau) e^{j2\pi\nu t} d\tau d\nu, \quad (2)$$

so the channel is a superposition of delay-and-frequency-shift operators. The other two Bello functions follow by Fourier transform in τ : the time-varying transfer function $L_H(t, f) = \int h(t, \tau) e^{-j2\pi f \tau} d\tau$ and the Doppler-variant transfer function.

Proposition 2.1 (Existence). *Equation (2) holds for any LTV operator whose kernel $h(t, \tau)$ is a tempered distribution in t ; the delay–Doppler representation is not a narrowband approximation but a change of coordinates on the kernel [16].*

This is the correction made to v0.1 of the spec, which had called delay–Doppler “the narrowband form”. What depends on the regime is not existence but *sparsity* (Proposition 2.2).

2.2 Delay–scale (wideband) representation

[P]

The same operator can be written as a superposition of delay-and-scale operators,

$$y(t) = \iint h_W(\tau, \alpha) \sqrt{\alpha} x(\alpha(t - \tau)) d\tau d\alpha, \quad \alpha > 0, \quad (3)$$

where h_W is the delay–scale (wideband) spreading function [1, 4]. A single reflector with radial closing speed v produces, in passband,

$$\alpha = \frac{c + v}{c - v} \approx 1 + \frac{2v}{c} \text{ (monostatic, two-way)}, \quad \alpha = 1 + \frac{v}{c} \text{ (one-way, passive)}. \quad (4)$$

Two conventions are pack-declared and recorded in the anchor’s **normalization** field rather than fixed by the core: the $\sqrt{\alpha}$ energy normalization, and whether the scaling is written $x(\alpha(t - \tau))$ as above or $x(\alpha t - \tau)$, which differ by $\tau \mapsto \alpha\tau$.

2.3 When is one representation sparse? The range-walk number

Consider one path with delay τ_0 and scale α_0 , observed over $t \in [0, T]$, acting on a passband signal $x(t) = u(t)e^{j2\pi f_c t}$ with envelope bandwidth B . In baseband the path is the operator

$$u \mapsto \sqrt{\alpha_0} u(\alpha_0(t - \tau_0)) e^{j2\pi\nu_0 t} e^{-j2\pi f_c \alpha_0 \tau_0}, \quad \nu_0 = (\alpha_0 - 1)f_c, \quad (5)$$

whose Bello kernel is $h(t, \tau) = \sqrt{\alpha_0} e^{j2\pi\nu_0 t} e^{-j2\pi f_c \alpha_0 \tau_0} \delta(\tau - \tau(t))$ with the *drifting delay*

$$\tau(t) = \alpha_0 \tau_0 + (1 - \alpha_0) t. \quad (6)$$

Proposition 2.2 (Range walk). *Let g be the delay-resolution kernel of the waveform (width $1/B$) and $\tilde{h} = h *_{\tau} g$ the resolution-limited kernel. Over an observation of length T , the resolution-limited spreading function $\tilde{S}_H(\tau, \nu)$ of the single path (5) has support of extent*

$$\Delta\tau \approx \max\left(\frac{1}{B}, |\alpha_0 - 1|T\right), \quad \Delta\nu \approx \max\left(\frac{1}{T}, |\alpha_0 - 1|B\right), \quad (7)$$

that is, about $\max(1, \Pi_s)$ resolution cells along each axis, where

$$\boxed{\Pi_s = TB|\alpha_0 - 1| = TBM \text{ (one-way)}} \quad (8)$$

is the range-walk number. In the delay-scale plane the same path is a single point (τ_0, α_0) at the resolution of the wideband ambiguity function.

Sketch. Write $\tilde{h}(t, \tau) = \sqrt{\alpha_0} e^{j2\pi\nu_0 t} g(\tau - \tau(t))$ up to a constant phase. Substituting $s = \tau - \tau(t)$ in the Fourier integral over $t \in [0, T]$,

$$\tilde{S}_H(\tau, \nu) = \frac{\sqrt{\alpha_0}}{\alpha_0 - 1} e^{-j2\pi(\nu - \nu_0)t^*(\tau)} G\left(\frac{\nu - \nu_0}{\alpha_0 - 1}\right) \Big|_{s \in \text{window}}, \quad t^*(\tau) = \frac{\tau - \alpha_0\tau_0}{1 - \alpha_0},$$

where G is the transform of g and has width B . The delay support is the set of τ for which $t^*(\tau) \in [0, T]$, of length $|\alpha_0 - 1|T$; the Doppler support is where G is non-negligible, of width $|\alpha_0 - 1|B$. When either is below one resolution cell the window and kernel widths take over. $\square$

Remark 2.3. Equation (8) is the Kelly–Wishner narrowband condition $TB(2v/c) \ll 1$ for the monostatic case [13], written as a dimensionless group. In the regime coordinates of Section 4, $\Pi_s = M \cdot TB = M\beta_B(f_c T)$. It is a property of the waveform and the motion jointly, so the threshold θ at which a pack switches canonical kind from `delay_doppler` to `delay_scale` is declared and tested by the pack, per regime box, and is not core logic.

Check 1. Is $\Pi_s = TB|\alpha - 1|$ the quantity you would use to decide between delay–Doppler and delay–scale processing, and is a threshold of order one (we have used $\theta \approx 0.1$ as a placeholder) the right order? For the passive wideband sonar case, is there structure in the modal picture that the delay–scale form loses at ranges you care about?

2.4 Realization versus statistic

[G] for the field, [P] for WSSUS

S_H describes one channel realization. If the channel is wide-sense stationary in t with uncorrelated scattering in τ (WSSUS), then

$$\mathbb{E}[S_H(\tau, \nu) S_H^*(\tau', \nu')] = C_H(\tau, \nu) \delta(\tau - \tau') \delta(\nu - \nu'), \quad (9)$$

and C_H is the *scattering function*. Its marginals are the power–delay profile and the Doppler power profile,

$$P(\tau) = \int C_H(\tau, \nu) d\nu, \quad P_\nu(\nu) = \int C_H(\tau, \nu) d\tau. \quad (10)$$

For a single realization the time-averaged delay profile is $\hat{P}_T(\tau) = \frac{1}{T} \int_0^T |h(t, \tau)|^2 dt$, which converges to P under WSSUS and ergodicity. The wideband analogue replaces (τ, ν) with (τ, α) [4]. Every anchor record states which of these it holds through `statistical_semantics` $\in \{\text{realization, expectation, estimate, posterior_samples, pseudo_label}\}$, and WSSUS, when an operation needs it, is a declared parameter `assume_wssus`, never a default.

Check 2. Are there operational cases where you would want a *local* scattering function (non-WSSUS, per 15) as a first-class kind, rather than a realization plus a time-series of PDPs?

3 Descriptors as functionals of a named representation

The anchor is a vocabulary, not a required payload: every quantity Crossfade estimates is defined as a functional of a *named* channel representation. In the contracts this is one string per quantity, `canonical_kind`, taking one of the values `impulse_response`, `delay_doppler`, `delay_scale`, `power_statistic` or `descriptors_only`, or else the tag `not_channel`. `[G]` that the string exists; `[P]` the list.

3.1 Moments of the power–delay profile

Definition 3.1 (Delay moments). For a non-negative profile P with $0 < \int P < \infty$,

$$\bar{\tau} = \frac{\int \tau P(\tau) d\tau}{\int P(\tau) d\tau}, \quad \sigma_\tau^2 = \frac{\int (\tau - \bar{\tau})^2 P(\tau) d\tau}{\int P(\tau) d\tau}. \quad (11)$$

For a discrete profile $\{(\tau_k, E_k)\}$ with weights $w_k = E_k / \sum_j E_j$: $\bar{\tau} = \sum_k w_k \tau_k$, $\sigma_\tau^2 = \sum_k w_k (\tau_k - \bar{\tau})^2$.

The Doppler spread σ_ν is the same functional of P_ν . Coherence bandwidth and coherence time are of order $1/\sigma_\tau$ and $1/\sigma_\nu$; the constant depends on the correlation level chosen and is a declared convention, not a physical fact.

Definition 3.2 (Resolvable-arrival count). `[P]` Given a delay resolution Δ and a relative threshold $\gamma \in (0, 1)$, cluster the arrivals of a discrete profile greedily in delay order, merging any arrival within Δ of the current cluster; let E_j be cluster energies. Then

$$N_{\text{arr}}(\Delta, \gamma) = \#\{j : E_j \geq \gamma \max_i E_i\}. \quad (12)$$

N_{arr} depends on (Δ, γ) and on the merging rule; two different rules are two different quantities, so the rule is recorded under `assumptions_used` and arrivals inside one cell are listed under `ambiguous`, never silently merged into a point estimate.

3.2 Exact descriptors from an arrival list

For a time-invariant channel given as an arrival list, $h(\tau) = \sum_k a_k \delta(\tau - \tau_k)$, the profile is $P(\tau) = \sum_k |a_k|^2 \delta(\tau - \tau_k)$ and (11) is exact with $E_k = |a_k|^2$. This is why W1’s simulator produces an arrival list: truth is the functional itself, not a second estimate.

Proposition 3.3 (Delay spread from pairwise differences). *With $w_k = |a_k|^2 / \sum_j |a_j|^2$,*

$$\sigma_\tau^2 = \frac{1}{2} \sum_{k,l} w_k w_l (\tau_k - \tau_l)^2. \quad (13)$$

Hence σ_τ is invariant to a common delay shift and is determined by the delay-difference profile $Q(\tau) = \sum_{k,l} |a_k|^2 |a_l|^2 \delta(\tau - (\tau_k - \tau_l))$ through $\sigma_\tau^2 = \frac{1}{2} \int \tau^2 Q / \int Q$.

Proof. Expand $\frac{1}{2} \sum_{k,l} w_k w_l (\tau_k - \tau_l)^2 = \sum_k w_k \tau_k^2 - (\sum_k w_k \tau_k)^2$ using $\sum_k w_k = 1$. $\square$

Remark 3.4 (Why this matters for a passive receiver). A receiver that does not know the source sees $y = s * h$ and can form the autocorrelation $R_{yy}(\tau) = \sum_{k,l} a_k a_l^* R_{ss}(\tau - (\tau_k - \tau_l))$. Its peaks sit at pairwise delay differences with weights $a_k a_l^*$, whose squared magnitudes are $|a_k|^2 |a_l|^2$, exactly the weights of Q . So an autocorrelation-based estimator can recover σ_τ without the absolute delay, *provided* the pairwise differences are resolved ($|\tau_k - \tau_l| > 1/B$ for $k \neq l$) and R_{ss} is narrow relative to them. When differences collide, the estimator is biased; W1 measures that bias against the exact value rather than assuming it away. The absolute delay $\bar{\tau}$ is not recoverable without a source reference, which is the delay ambiguity of Section 6.

Check 3. Do you agree that RMS delay spread (via the difference profile) and resolvable-arrival count (with a declared merging rule) are the right first two descriptors, and that they are what your cepstral and autocorrelation processing already estimate? If you would replace them, which functional of P or C_H , and why?

3.3 The T0 fixture, stated

The W0 workbench’s only computation is (11) on a supplied point-weight profile, plus a normalized width $\sigma_\tau/t_{\text{ref}}$. Golden case: delays $\{0, 2\}$ ms, weights $\{1, 1\}$, $t_{\text{ref}} = 2$ ms gives $\bar{\tau} = 1$ ms, $\sigma_\tau = 1$ ms, normalized width 0.5. It exists to exercise units, lineage, replay and refusal, not to estimate anything.

4 Regime coordinates

A regime coordinate is a dimensionless group that a task names as a candidate explanatory variable for a specific descriptor. The fields are **[G]**; whether any group predicts anything is **[H]**.

4.1 Candidate groups

[P]

Group	Definition	Governs	Placeholders (sonar; HF)
M	v/c	Doppler as shift vs. scale	$10^{-3}\text{--}10^{-2}$; $10^{-6}\text{--}10^{-5}$
β_B	B/f_c	sparsity with M , TB , eq. (8)	$0.3\text{--}1$; $10^{-3}\text{--}10^{-2}$
Π_τ	$\sigma_\tau B$	delay spread in resolution cells	$10\text{--}10^3$; $1\text{--}50$
Π_ν	$\sigma_\nu T$	Doppler spread in Doppler cells	$0.1\text{--}10$; $0.1\text{--}10$
Π_L	L/λ	aperture in wavelengths	$10\text{--}100$; $10\text{--}300$
Π_f	f/f_{cut}	propagating-mode count, eq. (16)	$1\text{--}50$; $1\text{--}10$
Π_h	$k h_{\text{rms}} \sin \theta$	coherent vs. diffuse boundary scatter	$0.1\text{--}10$; $0.1\text{--}5$
SNR_{cell}	per-resolution-cell SNR	detection-limited estimation	$-20\text{--}0$ dB; $0\text{--}30$ dB

Ranges are placeholders to be replaced by programme values. Some authors define the Rayleigh roughness parameter with a factor 2; the pack declares which.

4.2 The window question, made dimensionless

The transcript’s practical question, what FFT length and window to use for HF versus sonar, has one answer in both domains. For a short-time transform of window length T_w over a total observation T : arrivals separated by less than $1/B$ are not resolved, so the delay structure available to any estimator has Π_τ cells; the window must be short against the coherence time, $\sigma_\nu T_w \lesssim \varepsilon$, while the Doppler structure resolvable over the whole observation has Π_ν cells. Window length follows Π_ν ; resolution follows Π_τ . It is the same decision in both domains once written this way.

4.3 Per-descriptor governing sets and overlap **[P]** for the metric, **[H]** for the matching claim

Each descriptor q names a governing set G_q of groups. A dataset’s tested support $C_d(q)$ is a finite set of cells in $\log_{10}$ -coordinates over G_q . Define

$$O_q = C_{\text{src}}(q) \cap C_{\text{tgt}}(q), \quad d_q(c, C_{\text{src}}) = \min_{c' \in C_{\text{src}}(q)} \max_{g \in G_q} |\log_{10} \pi_g(c) - \log_{10} \pi_g(c')|, \quad (14)$$

so d_q is the number of decades a target cell sits from the nearest source cell in the worst governing group. The tested support is the list of evaluated cells, never a bounding box **[G]**.

Descriptor	G_q	Sonar–HF overlap under placeholders
RMS delay spread, arrival count	$\{\Pi_\tau, \Pi_f, \text{SNR}_{\text{cell}}\}$	partial
Interference structure, striation slope	$\{\Pi_f, \Pi_L, \Pi_\tau\}$	partial
Scale or Doppler spread	$\{M, \beta_B, \Pi_\nu\}$	none in M, β_B
Boundary scatter character	$\{\Pi_h\}$	partial

Proposition 4.1 (The overlap result). *Under the placeholder ranges, the sonar and HF intervals for M are separated by at least two decades and those for β_B by at least 1.5 decades. Hence for any descriptor with $M \in G_q$ or $\beta_B \in G_q$, $O_q = \emptyset$ and $d_q \geq 1.5$ for every target cell.*

Corollary 4.2 (Predicted negative control). *[H] If structure transfers only where governing groups overlap, Doppler and scale descriptors are predicted not to transfer between sonar and HF, and whatever transfers lives in $\{\Pi_\tau, \Pi_\nu, \Pi_f, \Pi_L\}$: arrival structure and waveguide interference. This is the transcript’s “ionosphere as the ocean flipped upside down” stated precisely: the shared object is the waveguide, not the Doppler. A measured gain on a Doppler descriptor would be a finding, not a failed gate.*

4.4 The mode-count coordinate

For the Pekeris waveguide of Section 5.1, mode m propagates above its cutoff

$$f_m = \frac{(2m-1)c}{4D\sqrt{1-(c/c_b)^2}}, \quad m = 1, 2, \dots, \quad (15)$$

so with $\Pi_f = f/f_1$ the number of propagating modes is

$$N_{\text{modes}}(f) = \left\lfloor \frac{2fD}{c} \sqrt{1-(c/c_b)^2} + \frac{1}{2} \right\rfloor = \left\lfloor \frac{\Pi_f + 1}{2} \right\rfloor. \quad (16)$$

In the ionosphere the sense is reversed: a hop mode propagates only *below* its maximum usable frequency, $f < f_{\text{MUF}} = f_{\text{crit}} \sec \varphi_0$ (flat earth, no field). A literal f/f_{cut} is therefore not the same quantity in both domains. What is comparable is the propagating-mode count itself, so we propose that each pack define Π_f such that N_{modes} is a monotone function of it, and record N_{modes} alongside.

Check 4. Is the propagating-mode count the right common coordinate for “modes above cutoff” across a fluid waveguide and the Earth–ionosphere duct, or would you define Π_f for HF as f_{MUF}/f , or something else?

Check 5. Does one “medium variability rate” (sound-speed-profile correlation time over T) belong in this list, or do internal waves and fronts need their own coordinate? Please replace the placeholder ranges with programme values where you can.

5 Generators

Both first generators are deliberately simple so that truth is exact by construction. Their limitations are stated; the sonar lead’s acceptance of the first is on the critical path (ADR-11).

5.1 Sonar: image-source Pekeris waveguide

[P]

Isovelocity water column of depth D , sound speed c , density ρ , over a fluid half-space (c_b, ρ_b) [11, 17]. Pressure-release surface at $z = 0$, bottom at $z = D$; source at $(0, z_s)$, receiver at (r, z_r) .

Images. Reflecting the source alternately in the surface ($z \mapsto -z$) and the bottom ($z \mapsto 2D - z$) generates image depths

$$z^{(n,\sigma)} = 2nD + \sigma z_s, \quad n \in \mathbb{Z}, \quad \sigma \in \{+1, -1\}, \quad (17)$$

with surface and bottom reflection counts

$$(n_s, n_b) = \begin{cases} (|n|, |n|) & \sigma = +1, \\ (n-1, n) & \sigma = -1, \quad n \geq 1, \\ (|n|+1, |n|) & \sigma = -1, \quad n \leq 0. \end{cases} \quad (18)$$

(Check: $(n, \sigma) = (0, +)$ is the direct path; $(0, -)$ one surface bounce; $(1, -)$ one bottom bounce; $(1, +)$ surface then bottom.)

Arrival list. For each image, path length, delay and grazing angle are

$$R = \sqrt{r^2 + (z_r - z^{(n,\sigma)})^2}, \quad \tau = R/c, \quad \theta = \arctan \frac{|z_r - z^{(n,\sigma)}|}{r}, \quad (19)$$

and the complex gain, with spherical spreading and a surface coefficient of -1 , is

$$a = \frac{(-1)^{n_s} [R_b(\theta)]^{n_b}}{R}, \quad R_b(\theta) = \frac{\rho_b c_b \sin \theta - \rho c \sin \theta_b}{\rho_b c_b \sin \theta + \rho c \sin \theta_b}, \quad \frac{\cos \theta}{c} = \frac{\cos \theta_b}{c_b}. \quad (20)$$

For $c_b > c$ there is a critical grazing angle $\theta_c = \arccos(c/c_b)$; below it $\sin \theta_b$ is imaginary, $|R_b| = 1$ and R_b carries a phase. That phase is frequency-independent for a lossless half-space, so it is applied to the analytic source signal:

$$y(t) = \Re \left\{ \sum_k a_k s_a(t - \tau_k) \right\} + n(t), \quad h(\tau) = \sum_k a_k \delta(\tau - \tau_k). \quad (21)$$

Truth for W1 is the list $\{(\tau_k, a_k, \theta_k, n_s, n_b)\}$; the descriptors of Section 3 are exact functionals of it. The sum is truncated when τ_k exceeds a declared maximum delay or $|a_k|$ falls below a declared floor; the truncation is recorded as a parameter.

What the model is and is not. With constant reflection coefficients the image sum is the exact solution for the ideal waveguide. With the angle-dependent $R_b(\theta)$ evaluated at each image's specular angle it is the ray (high-frequency) approximation to the Pekeris field: it omits the branch-line (head-wave) contribution and the modal corrections near cutoff, and it has no refraction, so no ducted paths. Mode count is nevertheless well defined through (15), which is how Π_f is assigned to a scene.

Check 6. Is this image model an acceptable first generator for delay-spread and arrival-count descriptors, with a refraction-capable normal-mode or ray code you already run as the second generator for the cross-domain step and the reality gate? Is applying $R_b(\theta)$ beyond the critical angle as a constant phase on the analytic signal how you would do it in the time domain?

5.2 HF: discrete-mode skywave over a Chapman layer [P]

Electron density in a Chapman layer with peak N_m at height h_m and scale height H (overhead sun),

$$N(h) = N_m \exp\left(\frac{1}{2}[1 - \zeta - e^{-\zeta}]\right), \quad \zeta = \frac{h - h_m}{H}, \quad f_p(h) [\text{Hz}] \approx 8.98 \sqrt{N(h) [\text{m}^{-3}]}, \quad (22)$$

critical frequency $f_{\text{crit}} = f_p(h_m)$ [6]. A wave incident on the layer at angle φ_0 from the vertical (flat earth, no magnetic field) reflects where $f \cos \varphi_0 = f_p(h)$, hence only if $f \cos \varphi_0 < f_{\text{crit}}$. Its

group delay is that of the equivalent triangular path reflected at the virtual height (Breit–Tuve and Martyn’s theorems),

$$h'(f_v) = \int_0^{h_r} \frac{dh}{\sqrt{1 - f_p^2(h)/f_v^2}}, \quad f_v = f \cos \varphi_0, \quad \tau_n = \frac{2n}{c} \sqrt{\left(\frac{R_g}{2n}\right)^2 + h'^2}, \quad (23)$$

for an n -hop mode over ground range R_g , with φ_0 found self-consistently from $\tan \varphi_0 = R_g/(2n h')$. Modes are enumerated over n and layers (E, F); the magnetoionic split is modelled as two modes with reflection conditions $f_p = f_v$ (ordinary) and $f_p^2 = f_v(f_v - f_H)$ (extraordinary), f_H the gyrofrequency. Per-mode gain combines spreading, a $1/f^2$ absorption law per D-region passage, and ground reflection on intermediate hops; ionospheric motion is a slow complex multiplicative process per mode with a declared coherence time. Truth is the mode list $\{(\tau_j, g_j(t))\}$, and the received signal has the same form as (21).

5.3 What the two generators share

They share no propagation code and different physics (boundary images in a fluid layer versus refraction in a plasma layer with ground hops). They share one approximation family: *discrete paths with per-path delay and gain*. That family is the anchor assumption itself, and the independence statement required for any cross-domain claim must say so [G]. Measured data is required before any real-world claim.

5.4 The waveguide invariant and its HF analogue

[H]

In a range-independent ocean waveguide the intensity at range r and frequency ω is a sum of modal cross terms $I(r, \omega) \propto \sum_{m,n} A_m A_n^* \cos[(k_m(\omega) - k_n(\omega))r]$. Along a striation the modal phase difference is constant, giving $d\omega/dr = \beta \omega/r$ with

$$\beta = -\frac{\Delta S_p}{\Delta S_g}, \quad S_p = \frac{k}{\omega}, \quad S_g = \frac{dk}{d\omega}, \quad (24)$$

the waveguide invariant [7]; $\beta \approx 1$ for low-order modes in a Pekeris waveguide, and it is estimable without range [5, 10, 18].

For any two discrete paths with delay difference $\Delta\tau(R)$ varying with a range-like parameter R , the interference $\cos[\omega \Delta\tau(R)]$ has striations along $d\omega/dR = -\omega (d\Delta\tau/dR)/\Delta\tau$, so the same construction defines

$$\beta_{\text{path}} = -\left(\frac{d \ln \Delta\tau}{d \ln R}\right)^{-1}, \quad (25)$$

which reduces to (24) for modal pairs. Hypothesis H4 asks whether β_{path} computed from HF hop-mode pairs over ground range is approximately constant across mode pairs and ranges, as β is for low ocean modes; if it is, the striation-slope descriptor is a mechanism that transfers, and if it is not, the analogy breaks at a named place.

Check 7. Is the flipped-ocean analogy strong enough that you would expect (25) to be approximately invariant for HF multimode returns? What would you look for first in a range–frequency display of an HF return?

6 Identifiability

6.1 The generative model

Each domain d is written as

$$y_d = \mathcal{M}_d[\mathcal{F}_d(\theta_{\text{sh}}, \theta_{\text{pr}}; c_d)] + \varepsilon_d, \quad (26)$$

with $\mathcal{F}_d$ the forward (propagation) model, $\mathcal{M}_d$ the measurement and processing chain, c_d the physical context (medium state, geometry, band), θ_{sh} the quantities the task treats as shared and θ_{pr} the rest. “Nuisance” is relative to a task, never intrinsic: each task declares which components of θ are retained, conditioned on, or marginalized $[\mathbf{G}]$. A change to $\mathcal{M}_d$ (resampling, windowing, magnitude-only) is lossy processing; a change to c_d (retune, sound-speed change) is a physical intervention; neither is a “convention”, and each transform a pack admits is classified as exact, coordinate, lossy or physical and tested individually $[\mathbf{G}]$.

6.2 Blind deconvolution

With $y = h * s$ and neither factor known, the following map y to itself:

$$(h, s) \mapsto (\kappa h, s/\kappa), \quad (h, s) \mapsto (h(\cdot - \tau_0), s(\cdot + \tau_0)), \quad (h, s) \mapsto (h * g, g^{-1} * s) \text{ for invertible } g. \quad (27)$$

Scale and delay are the minimal ambiguities; without structure, the whole family of invertible filters is. Identifiability up to scale requires structural constraints on h or s (subspace or sparsity), with conditions on the constraint dimensions relative to the observation length [14]. $[\mathbf{G}]$: a learned estimator does not remove an ambiguity, it hides it, so every estimator lists **assumptions**, **identifiable** and **ambiguous**, and a head that needs an ambiguous quantity returns alternatives or abstains.

6.3 What the classical estimators assume

Cepstrum. For $h = \sum_k a_k \delta(\tau - \tau_k)$ with a dominant first arrival, write $H(f) = a_0 e^{-j2\pi f \tau_0} \prod_{k \geq 1}$ factors; for two arrivals with $r = a_1/a_0$, $|r| < 1$, $\Delta = \tau_1 - \tau_0$,

$$\log(1 + r e^{-j2\pi f \Delta}) = \sum_{n \geq 1} \frac{(-1)^{n+1} r^n}{n} e^{-j2\pi f n \Delta}, \quad (28)$$

so the channel’s complex cepstrum is a train of impulses at quefrequencies $n\Delta$ with weights $(-1)^{n+1} r^n/n$, and $\log |Y| = \log |S| + \log |H|$ separates the source if $\log |S(f)|$ is smooth, meaning its cepstrum lies below the smallest Δ . Assumptions: $|r| < 1$ (the magnitude cepstrum cannot tell the ordering of two arrivals, only $|\Delta|$); arrivals separable in quefrecy ($\Delta > 1/B$); one source. Failure regimes follow: delay collisions, a non-smooth or non-stationary source spectrum, and overlapping sources, for which $Y = S_1 H_1 + S_2 H_2$ does not factor and the logarithm separates nothing.

Artificial time reversal. Beamforming an array toward one ray or mode arrival gives $P_b(f) \approx S(f) G_b(f)$ with $\arg G_b$ approximately linear in f ; then $\arg S(f) \approx \arg P_b(f) - 2\pi f \tau_b + \text{const}$, leaving an overall delay and scale unknown [19, 20]. Assumptions: known array geometry and local sound speed; a resolvable arrival with near-linear phase.

Bilinear models for sources of opportunity. If s lies in a known low-dimensional subspace and h in another, y is bilinear in the two coefficient vectors and the pair is recoverable up to the intrinsic ambiguities under dimension conditions [14, 22]. This is the acoustic template for the overlapped-source problem.

6.4 Multi-view identifiability, and what a deterministic view does not give

Let a recording be $x = f(\mathbf{c}, \mathbf{s})$ with content $\mathbf{c}$ and style $\mathbf{s}$, and let a second view $\tilde{x} = f(\mathbf{c}, \tilde{\mathbf{s}})$ share the content while the style is *independently* resampled. Under the conditions of von K  gelgen et al. [23] and Gresele et al. [8], a contrastive objective on $(x, \tilde{x})$ recovers $\mathbf{c}$ up to an invertible map (block identifiability).

Remark 6.1 (Deterministic re-representations give consistency, not isolation). If $\tilde{x} = \phi(x)$ for a deterministic ϕ (waveform to spectrogram, waveform to estimated CIR), then for any encoder e the pair $(e, e \circ \phi^{-1}|_{\text{range } \phi})$ attains perfect agreement, and no style is stripped because none varies. Such pairs test that an encoder is consistent across representations (what CSI-CLIP++ demonstrates under ideal ray-traced CSI, 12); they are not an alignment basis. Views with independent nuisance variation are: different sub-arrays on one event, different time windows of a stationary source through a slowly varying channel, different upstream chains on one recording, and controlled simulation pairs (same channel, different source; same source, different channel). This is the split between losses L2a and L2b.

6.5 The overlapped-source problem

Many sources in one band, $y = \sum_i h_i * s_i + n$, is blind multichannel separation. It becomes tractable only when the pack supplies structure that makes it identifiable: a source subspace (tonals plus a smooth continuum), the array manifold, and sparsity of each source’s arrival set. An encoder does not discover this; it operates inside it.

Check 8. In the overlapped-source case, what specifically breaks the cepstral method for you: delay collisions, source non-stationarity, or association? Is there a source-subspace assumption that would make it identifiable in the bilinear sense of Tian et al. [22]? Below what Π_τ is arrival sparsity no longer a usable prior?

7 Transfer as a measured quantity

7.1 Losses, budgets, effect

[G]

For a task with descriptor q , an arm a trained with N target labels, and independent units $u \in U$ (scenes, sessions, instruments; never windows), the loss is aggregated at the unit,

$$L_a(N) = \frac{1}{|U|} \sum_{u \in U} \ell(\hat{q}_{a,u}, q_u), \quad \ell = |\hat{\sigma}_\tau - \sigma_\tau| B \text{ for delay spread (error in resolution cells).} \quad (29)$$

The effect of a candidate over a baseline is

$$\delta(N) = L_{\text{base}}(N) - L_{\text{cand}}(N), \quad (30)$$

positive favouring the candidate, with a confidence interval from bootstrap over units and over training seeds. No ratio undefined at zero baseline loss is used.

7.2 Hypotheses

[H]

Hypothesis 7.1 (H1, transfer at overlap). For fixed q and N , on cells in O_q , $\delta(N) = L_{\text{scratch}}(N) - L_{\text{pretrained}}(N)$ exceeds a preregistered meaningful δ_{min} with its CI excluding zero.

Hypothesis 7.2 (H2, transfer surface). δ is a decreasing function of $d_q(c, C_{\text{src}})$ in (14). Reported as the surface $\delta(c)$ against d_q ; a gain at large d_q is a finding, never a failed gate.

Hypothesis 7.3 (H3, regime conditioning). Conditioning on regime coordinates improves held-out utility over unconditioned arms under interventions that separate regime from domain identity. Routing telemetry alone is not evidence.

Hypothesis 7.4 (H4, mechanism). β_{path} of (25) estimated in one bounded dispersive waveguide remains predictive in another and beats shuffled-pair and nuisance-matched controls.

Hypothesis 7.5 (H5, uncertainty under shift). Coverage $\text{cov}(c) = \frac{1}{n_c} \sum_{u \in c} \mathbb{I}[q_u \in I_u]$ at nominal level survives shift across cells, or the applicability flag marks the cells where it does not. Reported with interval width, accepted-case error and abstention rate, each with its denominator.

7.3 The baseline ladder as nested comparisons

[G] for the order

Arms in order: (1) classical or direct calculation; (2) physically normalized features plus a small head; (3) independent target model from scratch; (4) target-only self-supervised pretraining; (5) pooled model with a domain token; (6) dense shared/private with adapters; (7) regime-routed experts, optional. Every arm above the first is compared to every arm below it at declared budgets. Arm 3 is mandatory because physics foundation models are conditional generalists with negative transfer in a substantial fraction of cases [3], and arm 7 is an ablation because its motivating result covers two fluid regimes [21]. Well-tuned simple baselines are hard to beat [9].

7.4 Decision rule and what is not evidence

[G]

$\delta_{\min}$ is set from the from-scratch arm’s development-set variability across seeds and frozen with a timestamp before the locked test partition is read. Supported: the CI of δ excludes zero and exceeds $\delta_{\min}$ at two or more budgets. Evidence against: the CI excludes $\delta_{\min}$ in the other direction at two or more budgets. Otherwise inconclusive. A crash is not negative transfer; lack of significance is not evidence of no effect; a result on one simulator relabelled is not cross-domain evidence.

7.5 Fusion, two rules only

[G]

Let $\mathcal{R}(r)$ be the set of root records a result r derives from. If $\mathcal{R}(r_1) \cap \mathcal{R}(r_2) \neq \emptyset$, the two are one piece of evidence and the more conservative (wider) is kept; they are never multiplied. If the model-error dependence between r_1 and r_2 is unknown, they are reported side by side and not combined numerically. Any further rule needs its own association contract, dependence model and calibration test.

Check 9. The pilot is designed so that arrival-structure descriptors have overlapping cells and Doppler descriptors do not, and it measures δ against d_q per descriptor rather than demanding a yes or no. Does that design lose anything you valued in the original conversation? Is descriptor-matched pairing across the two simulators an acceptable primary pairing basis, or do you want pairs defined from a shared physical scene?

The checks, collected

For convenience, the numbered checks above, in the order they appear:

- Check 1.** The range-walk number $\Pi_s = TB|\alpha - 1|$ as the delay-Doppler versus delay-scale criterion, and its threshold.
- Check 2.** Whether a local (non-WSSUS) scattering function is needed as a first-class kind.
- Check 3. Critical path.** RMS delay spread and resolvable-arrival count as the first two descriptors.
- Check 4.** The propagating-mode count as the common “modes above cutoff” coordinate across ocean and ionosphere.
- Check 5.** Whether medium variability needs its own coordinate; programme ranges for the groups.
- Check 6. Critical path.** The image-source Pekeris generator as the first generator, and the beyond-critical-angle treatment.
- Check 7.** Whether β_{path} should be expected to be invariant in HF multimode returns.
- Check 8.** What breaks the cepstral method with overlapped sources, and whether a source subspace makes it identifiable.
- Check 9.** Whether the per-descriptor pilot design loses anything, and the pairing basis.

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